NETWORKING FUNDAMENTALS III


TEKsystems
Global Services
WORKFORCE
DEVELOPMENT

COURSE STRUCTURE

Day 1: Networking Fundamentals I

Day 2: Networking Fundamentals II

Day 3: Networking Fundamentals II

Day 4: Network Security in Depth

Day 5: Routing Fundamentals & Capstone

AGENDA FOR DAY 3

1. Systematic Troubleshooting Methodology
2. OSI-Based Troubleshooting
3. Essential Diagnostic Tools: ping, traceroute, Wireshark
4. Cisco IOS Verification Commands
5. Course Review & Knowledge Check
6. Bridge to Routing & Switching Essentials

TROUBLESHOOTING METHODOLOGY

THE STRUCTURED TROUBLESHOOTING APPROACH

Step 1 — Define the problem: exactly what is broken? What was the last change made?

Step 2 — Gather information: ping tests, show commands, interface counters, logs, topology

Step 3 — Analyze: identify the likely layer and form a hypothesis

Step 4 — Test: make ONE change at a time, then retest — never change multiple things simultaneously

Step 5 — Eliminate: if the change didn't fix it, revert and try the next hypothesis

Step 6 — Document: record root cause and fix — builds the team's knowledge base

Top-down (L7→L1) vs Bottom-up (L1→L7): choose based on symptoms — no link light = start at L1

OSI-BASED TROUBLESHOOTING

- Layer 1 (Physical): check link lights, cable type, damaged cable, correct SFP/transceiver, speed/duplex
- Layer 2 (Data Link): verify MAC table, VLAN assignment, trunk config, STP port state, CRC/FCS errors
- Layer 3 (Network): verify IP addressing, subnet mask, default gateway, routing table, ACLs applied
- Layer 4 (Transport): confirm correct port is open, firewall rules, NAT translation table
- Layer 5–7 (Application): test with curl/browser, DNS resolution, TLS certificate validity, app logs
- Strategy: no connectivity at all → start at L1; partial connectivity → start at L3–4

POP QUIZ:

A user can ping their default gateway but not reach any external websites. What should you check first?

A. The user's NIC driver
B. DNS resolution and routing beyond the gateway
C. The user's operating system
D. Switch port VLAN assignment

POP QUIZ:

A user can ping their default gateway but not reach any external websites. What should you check first?

A. The user's NIC driver
B. **DNS resolution and routing beyond the gateway**
C. The user's operating system
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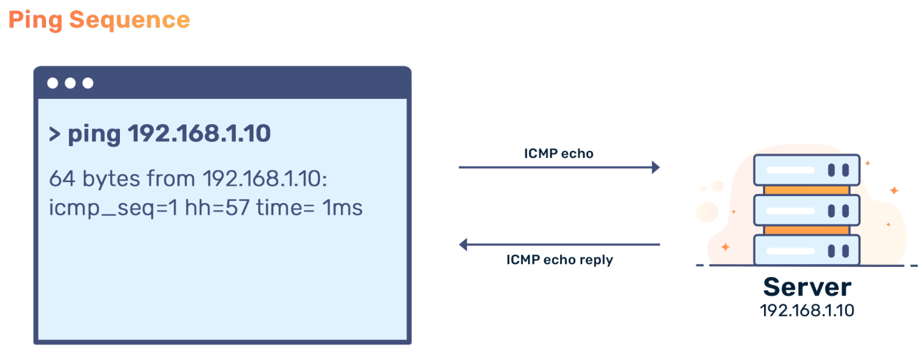
ESSENTIAL DIAGNOSTIC TOOLS

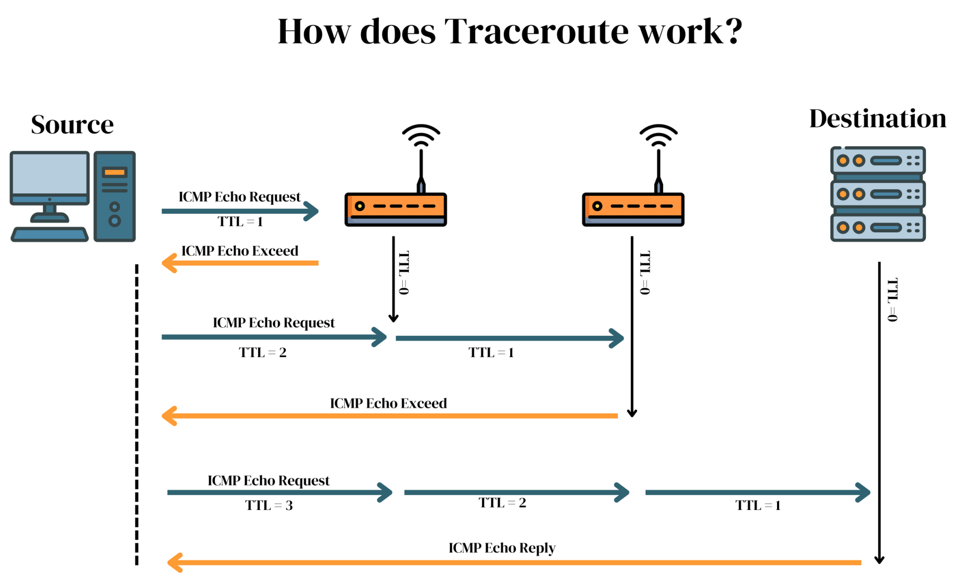
PING — TESTING REACHABILITY LAYER BY LAYER

- ping 127.0.0.1 — tests TCP/IP stack on local machine (loopback)
- ping <default-gateway> — confirms Layer 3 connectivity to the router
- ping <remote-host> — tests end-to-end path through the network
- Extended ping (Cisco IOS): specify source interface, count, size, timeout
- Interpret: ! = reply | . = timeout | U = unreachable | M = MTU issue
- If gateway ping fails: L1/L2 issue (cable, VLAN, ARP) — check before assuming L3
- If gateway succeeds but remote fails: routing or ACL issue beyond the gateway



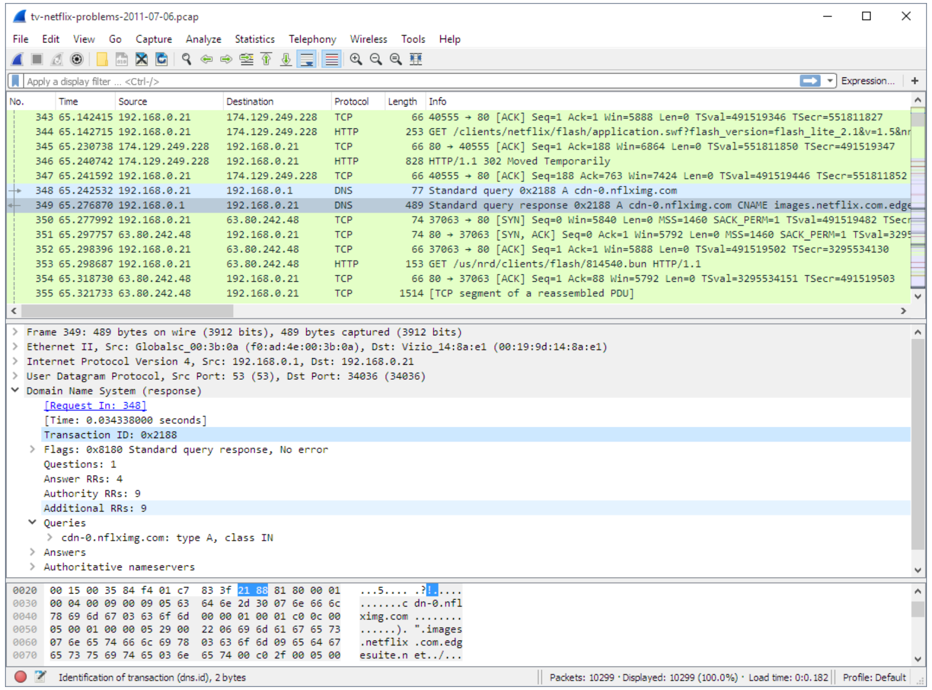
TRACEROUTE — MAPPING THE NETWORK PATH

- Sends packets with incrementing TTL; each router sends back 'Time Exceeded' revealing its IP
- Output: hop number | IP address | round-trip time (3 probes per hop)
- Asterisks (***): router exists but filters ICMP — not necessarily broken
- Identifies exactly WHERE packets are being dropped or excessively delayed
- Windows: tracert (uses ICMP) | Linux/Mac: traceroute (uses UDP by default)
- traceroute -I on Linux forces ICMP — better behavior against Cisco devices
- Useful for: routing loops, suboptimal paths, ISP handoff point verification



WIRESHARK — PACKET CAPTURE & ANALYSIS

- Captures and displays every packet on a network interface in real time
- Color-coded by protocol; shows all layers: Ethernet, IP, TCP/UDP, DNS, HTTP, TLS
- Filter syntax: ip.addr==192.168.1.1 | tcp.port==80 | dns | arp | icmp | http
- Follow TCP Stream: right-click → Follow → TCP Stream — see full HTTP conversation
- ARP analysis: spot duplicate IPs, gratuitous ARPs, ARP poisoning attempts
- DNS: verify correct resolution, check response IPs and TTL values
- Best for: unexplained drops, application debugging, security incident analysis



CISCO IOS VERIFICATION COMMANDS

- show ip interface brief — all interfaces, IPs, status (up/up, admin down, down/down)
- show running-config | section interface — show all interface configs
- show ip route — routing table; C=connected S=static O=OSPF D=EIGRP
- show arp — ARP cache (IP↔️MAC mappings on this device)
- show mac address-table — switch MAC table: which MAC is on which port/VLAN
- show vlan brief — VLAN database and port assignments
- show interfaces — full statistics: CRC errors, input/output drops, runts, giants
- Pipe modifiers: | include, | section, | begin — filter large outputs

ARP, ICMP & MTU IN PRACTICE

ARP (Address Resolution Protocol): resolves IP → MAC within local subnet

arp -a (Windows/Linux): shows local ARP cache — useful to verify L2 discovery

Gratuitous ARP: device announces its own MAC/IP — used by HSRP, causes cache refresh

ICMP TTL: each router decrements TTL by 1; at 0 → Time Exceeded sent back to sender

MTU (Maximum Transmission Unit): Ethernet = 1500 bytes; oversized packets fragmented or dropped

Path MTU Discovery: TCP uses 'don't fragment' bit; routers return 'Fragmentation Needed' if packet too large

Jumbo frames: MTU > 1500 (up to 9000) — used in data center storage traffic

POP QUIZ:

In a traceroute output, what do three asterisks (***) for a hop typically indicate?

A. The hop does not exist
B. ICMP is filtered or the router is not responding to probes
C. The path has a routing loop
D. The destination is unreachable

POP QUIZ:

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DAY 1 RECAP: FOUNDATIONS & ADDRESSING

- Networks connect devices via endpoints, links, and intermediaries — scales from home to Internet
- OSI 7-layer model gives us a common language for design and troubleshooting
- TCP = reliable, ordered (SYN/SYN-ACK/ACK); UDP = fast, connectionless
- IPv4: 32-bit dotted decimal; RFC1918 private ranges; CIDR /N notation
- Subnetting: block size = 256 - mask octet; VLSM assigns different sizes to match requirements
- IPv6: 128-bit hex notation; link-local (fe80::/10) auto-assigned; no broadcast

DAY 2 RECAP: SERVICES & SECURITY

- DNS: hierarchical name resolution — A, AAAA, CNAME, MX, PTR, TXT records; UDP/TCP 53
- DHCP DORA: Discover → Offer → Request → Acknowledge; relay agent crosses routers
- NAT/PAT: private IPs hidden behind one public IP using port numbers
- ACLs: standard (source IP only) applied near destination; extended applied near source
- Firewalls: stateless (packet filter) vs stateful (connection tracking) vs NGFW
- Wi-Fi: WPA3 > WPA2 > WPA (avoid) > WEP (never); enterprise uses 802.1X + RADIUS
- VPNs: IPsec (site-to-site) and SSL/TLS (remote access) encrypt over untrusted networks

POP QUIZ:

Which command quickly shows all interfaces, their IP addresses, and status on a Cisco device?

A. show running-config
B. show ip route
C. show ip interface brief
D. show arp

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POP QUIZ:

Which OSI layer does a switch primarily operate at?

A. Layer 1 – Physical
B. Layer 2 – Data Link
C. Layer 3 – Network
D. Layer 4 – Transport

WHAT YOU'VE BUILT — AND WHAT'S NEXT

- You now understand how data moves across any network — from bits on a wire to application data
- These fundamentals are the foundation every network engineer builds on every day
- Routing & Switching Essentials takes every topic deeper into enterprise-grade configuration
- In R&S you will configure: VLANs, trunks, STP, OSPF, EIGRP, HSRP, EtherChannel, data center design
- The Cisco IOS commands you learned here become your daily toolkit in R&S labs
- Cloud networking courses map these exact concepts to AWS/Azure (VPC=subnet, SG=firewall, Route Table=routing)

LAB 07: NETWORK DIAGNOSTICS & VERIFICATION

- Goal: Build a multi-subnet network and verify every layer using Cisco IOS diagnostic tools
  - Build: 1 router, 2 switches, 4 PCs, 1 server — two LAN subnets (192.168.10.0/24 and 192.168.20.0/24)
  - Configure: Static IPs on all devices; both router interfaces up with correct addressing
  - Ping layer by layer: loopback → gateway → same subnet → remote subnet
  - Traceroute: Map the path LAN1 → LAN2; interpret hops, RTT, and ***
  - Inspect Layer 2: show arp and show mac address-table — verify IP-to-MAC resolution
  - Verify: show ip interface brief and show ip route — confirm interfaces and routing table
  - Error counters: show interfaces — identify CRC errors, runts, and drops
  - Filter outputs: Practice | include, | section, | begin
  - Document: OSI layer summary table — tool, command, and finding at each layer

LAB 08: BUILD & VERIFY A COMPLETE SMALL NETWORK

- Goal: Apply all 3 days of learning to design and implement a working office network
- Steps:
  - Design: 192.168.0.0/24 split into 3 subnets: 60 hosts, 25 hosts, WAN /30
  - Build in Packet Tracer: 1 router, 2 switches, PCs, 1 server (DNS+DHCP)
  - Configure DHCP scopes for both LAN subnets; DNS server with A records
  - Configure PAT for Internet access; static NAT for the server (inbound access)
  - Apply ACL: deny Telnet from LAN2 to LAN1; permit everything else
  - Verify: all PCs get IPs, resolve DNS names, ACL enforced, router has correct routing table
  - Document: IP plan table, network diagram, list of CLI commands used